EXPANDED MASTER FIELD REFERENCE: Canned Motor & Wet-Rotor Circulators (Part 1)

General Architectural Rules

- **Cast Iron (F/CI):** Closed-loop hydronic heating/cooling *only*. Never use for potable water.
- **Stainless Steel (SF/SS) / Bronze (B):** Mandatory for open-loop and domestic hot water (DHW).
- **ECM (Electronically Commutated Motor):** High-efficiency, variable-speed magnetic motors. These auto-adapt to system pressure (ΔP-V or ΔP-C) and significantly reduce amp draw compared to traditional PSC (Permanent Split Capacitor) motors.

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1. TACO 00-SERIES & 00e-SERIES (ECM)

Architecture: Field-replaceable cartridge design. The volute remains piped during motor/cartridge replacement.

- **F** = Cast Iron | **SF** = Stainless Steel | **B** = Bronze | **IFC** = Integral Flow Check

Model	Motor	HP	Max Amps	Max GPM	Max Head (ft.)	Primary Application
003	PSC	1/40	0.45	7	4.5	DHW recirculation, very small loops
005	PSC	1/35	0.54	19	9	Small zoning, light hydronic heating
006	PSC	1/40	0.52	10	9	DHW recirculation ("Plumb n' Plug")

007	PSC	1/25	0.71	23	10	The standard residential hydronic workhorse
008	PSC	1/25	0.84	14	15	Radiant mixing, higher head/lower flow loops
009	PSC	1/8	1.40	10	35	High head drops (solar, heat recovery loops)
0010	PSC	1/8	1.10	30	10	High flow / low head multi-speed
0011	PSC	1/8	1.76	31	31	High head / medium flow boilers
0012	PSC	1/8	1.33	50	14	Very high flow residential / light commercial
0013	PSC	1/6	2.00	34	33	High head / high flow commercial
0014	PSC	1/8	1.45	32	22	Medium-high flow and head
0015	PSC	1/20	0.75	18	17	3-Speed universal replacement pump
007e	ECM	Var.	0.44	16	10	High-efficiency 007 replacement

0015e3	ECM	Var.	0.44	16	18	3-setting ECM replacement for UPS15-58
0018e	ECM	Var.	0.44	16	18	Bluetooth-enabled ECM for precise zoning
0034E	ECM	Var.	1.42	34	34	High-performance commercial ECM

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2. GRUNDFOS UP / UPS / ALPHA / MAGNA SERIES

Architecture: Aluminum oxide ceramic shaft and radial bearings.

- **UP** = Single speed | **UPS** = 3-speed | **ALPHA/MAGNA** = ECM Variable Speed | **F** = Flanged | **B/N** = Bronze/Stainless

Model	Speeds/Type	Max Amps	Max GPM	Max Head (ft.)	Primary Application
UP 15-10	1-Speed	0.22	8	5.5	DHW recirculation
UP 15-42	1-Speed	0.74	22	15	Standard residential hydronic heating
UPS 15-58	3-Speed	0.75	17	19.5	Universal 3-speed replacement
UP 26-96	1-Speed	1.70	28	30	High head applications (solar/heat exchangers)

UP 26-99	1-Speed	2.15	34	32	High flow / high head
UPS 26-150	3-Speed	1.70	52	45	Large residential / light commercial
UP 43-75	1-Speed	2.15	42	25	High flow light commercial
UPS 43-100	3-Speed	3.50	65	31	Medium commercial hydronics
ALPHA1 15-55	ECM	0.65	12	19	Standard high-efficiency auto-adapt
ALPHA2 15-55	ECM	0.65	21	19	Premium ECM w/ digital interface
MAGNA1	ECM	Var.	Up to 302	Up to 57	Commercial variable speed (Multi-size range)
MAGNA3	ECM	Var.	Up to 329	Up to 62	Premium commercial building management ECM

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3. BELL & GOSSETT (XYLEM) NRF / NBF / ECOCIRC

Architecture: DuraGlide bearing system with a self-cleaning particle shield. NRF (Cast Iron), NBF (Bronze), SSF (Stainless). ecocirc lines feature ultra-efficient spherical motor technology.

Model	Type	HP/Watts	Max GPM	Max Head (ft.)	Primary Application

NRF-22	1-Speed	1/25 HP	22	15	Standard residential zoning
NRF-25	3-Speed	1/15 HP	16	16	3-Speed residential replacement
NRF-33	3-Speed	1/15 HP	26	33	High head (radiant manifolds)
NRF-36	3-Speed	1/6 HP	36	24	High flow / medium head
NRF-45	3-Speed	1/6 HP	45	35	Light commercial, heavy residential
NBF-22	1-Speed	1/25 HP	22	15	Bronze DHW equivalent to NRF-22
ecocirc 19-16	ECM	4-65 W	19	16	Smart residential ECM (Proportional/Constant)
ecocirc 20-18	ECM	4-70 W	18	20	Heavy residential ECM w/ eAdapt auto-learn
ecocirc XL	ECM Com.	Up to 2 HP	Up to 275	Up to 65	Large commercial ECM line (Multi-size range)
PL Series	Dry-Motor	1/12 - 1/6	Up to 40	Up to 35	Maintenance-free dry motor high-performance

EXPANDED MASTER FIELD REFERENCE: Circulators (Part 2)

IV. WILO

Wilo is a heavy hitter in both the OEM replacement market and high-efficiency commercial spaces.

- **Architecture:** Star series utilizes quick-connect spring-clip wiring terminals (no wire nuts needed). Stratos series utilizes ultra-efficient ECM (Electronically Commutated Motor) technology.
- **Nomenclature: Star S** = Standard 3-speed cast iron | **Star Z** = Bronze/Stainless DHW | **FX** = Flanged + Check Valve

Performance Matrix (Standard 115V, 1Ph, 60Hz unless noted)

Model	Type	Max Amps / Watts	Max GPM	Max Head (ft.)	Primary Application
Star S 16	3-Speed PSC	0.54 A	16	16	Baseboard/radiator loops, small zones
Star S 21	3-Speed PSC	0.97 A	21	21	Standard multi-purpose hydronics
Star S 33	3-Speed PSC	1.80 A	33	33	High flow / high head, large radiant loops
Stratos ECO	ECM Auto	2.0 - 3.5 W	15.5	16	Small residential high-efficiency replacement
Stratos MAXO	ECM Smart	Var. (Up to 1.5 HP)	Up to 550	Up to 167	Premium commercial building management (Multi-size)

V. ARMSTRONG FLUID TECHNOLOGY

Armstrong's circulator line directly competes with Grundfos and Taco in the residential hydronic space, while their Compass line pushes into the high-efficiency smart pump sector.

- **Architecture:** Astro 2 features e-coated cast iron internals to resist corrosion and a self-venting design.
- **Nomenclature: CI** = Cast Iron (Closed Loop) | **SS / SSU** = Stainless Steel / Union Connection (Open Loop)

Performance Matrix (Standard 115V, 1Ph, 60Hz)

Model	Type	Max Amps / Power	Max GPM	Max Head (ft.)	Primary Application
Astro 230	3-Speed PSC	0.75 A	18	17	Standard residential replacement (Competes w/ UPS15-58)
Astro 250	3-Speed PSC	1.40 A	24	24	Medium flow / medium head hydronics
Astro 280	3-Speed PSC	1.95 A	36 (up to 64)	30 (up to 42)	High flow / high head
Compass H 20-20	ECM Wet-Rotor	5 - 45 W	20	20	Universal residential ECM replacement
Compass R	ECM Dry-Rotor	1/6 to 1/2 HP	Up to 225	Up to 65	Commercial variable speed (Multi-size range)

VI. BELL & GOSSETT (XYLEM) INLINE SERIES

Note: While the previous sections covered "wet-rotor" (canned) pumps, no field guide is complete without B&G's dry-motor inline pumps, which are the industry standard for older residential boilers and modern commercial hydronics.

- **Architecture:** These feature mechanical seals and separate motors. The Series 100 is famous for its oil-lubricated sleeve bearing and spring-coupler (repairable in the field). The e-60 and e-90 are modern, maintenance-free heavy-duty equivalents.

Performance Matrix (Standard 115V or Multi-Voltage)

Model	Type	HP Range	Max GPM	Max Head (ft.)	Primary Application
Series 100	3-Piece Oil Lubed	1/12 HP (1.75A)	33	8	The classic residential boiler circulator
Series PR	3-Piece Oil Lubed	1/6 HP (1.90A)	38	17	Upgraded residential / light commercial
Series e-60	Flex-Coupled Inline	1/4 to 3 HP	Up to ~150	Up to ~75	Medium-to-heavy commercial HVAC transfer
Series e-90	Close-Coupled Inline	1/4 to 5 HP	Up to ~250	Up to 200+	High pressure commercial building hydronics

EXPANDED MASTER FIELD REFERENCE: Circulators (Part 3)

VII. LAING THERMOTECH (A Xylem Brand)

Laing specializes almost exclusively in high-efficiency, spherical-motor pumps. They are heavily utilized in DHW recirculation, under-sink instant hot water systems, and solar thermal applications.

- **Architecture:** Spherical motor design with a single moving part (the hemispherical rotor/impeller unit). It sits on an ultra-hard ceramic ball. This design prevents rotor seizure from lime scale buildup.

- **Nomenclature: E** = ECM/Variable | **D5** = Solar/DC | **AutoCirc** = Under-sink DHW kit.

Performance Matrix (Standard 115V or 12-24V DC for Solar)

Model	Type	Power	Max GPM	Max Head (ft.)	Primary Application
E1	ECM Spherical	10 W	4.5	4	Tiny DHW loops / AutoCirc under-sink kits
E3	ECM Spherical	28 W	11	10	Standard DHW recirculation
E5	ECM Spherical	45 W	16	16	Large DHW loops / light radiant
D5 Solar	DC Brushless	Var. (DC)	6	14	Off-grid solar thermal / 12V-24V systems

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VIII. AQUAMOTION

A rising US-based competitor that specifically targets the replacement market for Taco and Grundfos, offering highly competitive standard hydronic and DHW pumps.

- **Architecture:** Features replaceable flushable cartridges and highly efficient PSC and ECM motors. Known for their "Aqua-Shield" outdoor DHW pumps.
- **Nomenclature: AM** = AquaMotion | **F** = Flanged (Iron) | **SF / S** = Stainless | **R** = Recirculation

Performance Matrix (Standard 115V, 1Ph, 60Hz)

Model	Type	Max Amps	Max GPM	Max Head (ft.)	Primary Application

AM7	1-Speed PSC	0.54	20	11	Direct Taco 007 / Grundfos UP15-42 replacement
AM5	1-Speed PSC	0.54	18	10	Light hydronic heating / zoning
AM10	3-Speed PSC	0.75	16	20	3-Speed replacement (Competes w/ UPS15-58)
AMRe / AMSe	ECM Variable	0.44	16	18	High-efficiency heating & DHW
O-Bypass	DHW Kit	Var.	3 to 5	5 to 10	Under-sink dedicated return line kits

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IX. TACO & GRUNDFOS INLINE COMPETITORS (Commercial/Heavy Duty)

To complete the picture started by Bell & Gossett's dry-motor Series 100 and e-60 in Part 2, here are the direct equivalents from Taco and Grundfos used in older residential boilers and medium-to-large commercial buildings.

- **Architecture:** Dry-motor, mechanical shaft seals. Designed to be rebuilt in the field (motors, couplers, bearing assemblies, and seals can all be swapped without removing the volute).

Performance Matrix (Standard 115V, or Multi-Voltage Commercial)

Brand / Model	Type	HP Range	Max GPM	Max Head (ft.)	Primary Application
Taco 110 Series	3-Piece Oil Lubed	1/12 to 1/8	Up to 60	Up to 15	Direct competitor to B&G Series 100

Taco 1600 Series	In-Line Base Mount	1/4 to 3 HP	Up to 220	Up to 70	Commercial HVAC / Chilled Water
Taco 1900 Series	Close-Coupl ed	1/4 to 2 HP	Up to 160	Up to 70	Compact commercial / competing w/ B&G e-90
Grundfos TP	Vertical In-Line	1/2 to 100+ HP	Up to 4,000+	Up to 400+	Heavy commercial building hydronics / cooling

EXPANDED MASTER FIELD REFERENCE: Circulators (Part 4)

X. KSB (Formerly SMEDEGAARD)

Smedegaard was a highly respected European pump manufacturer. They have since been absorbed by KSB, which has integrated their technologies into the KSB lineup.

- **Field Note:** If replacing an old Smedegaard, check the flange-to-flange (port-to-port) distance. Many can be directly replaced by the **KSB Calio** series, or standard Grundfos MAGNA / Wilo Stratos equivalents.
- **Current Lineup:** KSB's primary wet-rotor hydronic circulator line is the **Calio** (and Calio S for smaller residential applications). They are high-efficiency ECM pumps.

Performance Matrix (KSB Calio Series - ECM)

Model	Type	Max Power	Max GPM	Max Head (ft.)	Primary Application
Calio S	ECM Smart	Up to 90 W	Up to ~15	Up to ~25	Small residential hydronics / Smedegaard EV replacement
Calio	ECM Smart	Up to 1.5 kW	Up to ~220	Up to ~60	Commercial building management / Smedegaard Omega replacement

XI. OEM-SPECIFIC BOILER PUMPS (Internal)

Modern high-efficiency modulating-condensing boilers (e.g., Navien, Lochinvar, Viessmann, Triangle Tube) often use proprietary pumps built directly into the cabinet.

- **The Catch:** While the internal motor heads are usually manufactured by Wilo or Grundfos (e.g., a Wilo Para or Grundfos UPM3), they use **custom volutes** and **proprietary PWM (Pulse Width Modulation) wiring harnesses**.
- **Field Rule:** You generally *cannot* swap these with off-the-shelf pumps from a supply house. If the pump fails, you must order the exact OEM replacement part number from the boiler manufacturer to ensure the boiler's circuit board can properly communicate with it.

XII. GEOTHERMAL FLOW CENTERS

Ground-source heat pumps require specialized flow centers to push heavy glycol mixtures through hundreds of feet of buried earth loops.

- **Architecture:** These are either pressurized or non-pressurized canisters housing one or two high-head circulators.
- **Common Brands:** Geo-Flo, B&G (Geo series), Taco (Geo-Trak).
- **Pump Heads:** The actual pump motors on these flow centers are almost always specialized, heavy-duty versions of the **Grundfos UP26-99** or **Taco 0026 / 0032** series.

XIII. DOMESTIC PRESSURE BOOSTERS (Compact)

While technically not "circulators" (they boost open-system line pressure rather than circulating fluid in a closed loop), service technicians encounter them constantly.

- **Architecture:** Typically a multistage centrifugal pump paired with an internal diaphragm tank, pressure sensor, and flow switch, all in one compact housing.

Performance Matrix (Standard 115V or 230V)

Brand / Model	Type	HP	Max GPM	Max Boost (PSI)	Primary Application
Grundfos SCALA2	VFD / Constant Pressure	3/4	14	Up to 65	Whole-house residential city/well water boosting

Grundfos MQ	Standard Pressure Switch	3/4 or 1	18	Up to 65	Classic heavy-duty residential booster (Phasing out for SCALA)
Davey BT Series	Torrium2 Controller	1/2 to 1+	Up to 30	Up to 70	High-performance boosting and rainwater harvesting

EXPANDED MASTER FIELD REFERENCE: Circulators (Part 5 - Global & European Heavyweights)

XIV. DAB PUMPS

DAB (an Italian manufacturer) is a massive global player, highly prevalent in Europe and increasingly common in international commercial systems and packaged skids.

- **Architecture:** They heavily utilize high-efficiency synchronous motors with built-in frequency converters.
- **Nomenclature: Evosta** = Residential/Light Commercial (direct competitor to Grundfos Alpha / Wilo Yonos). **Evoplus** = Medium to Heavy Commercial (competitor to MAGNA3 / Stratos MAXO). **SAN** = Sanitary/Bronze DHW.

Performance Matrix (Standard 230V, 50/60Hz)

Model	Type	Max Power	Max Flow	Max Head	Primary Application
Evosta 2 (40-70/130)	ECM Smart	35 W	3.6 m³/h	6.9 m	Universal residential hydronic replacement
Evosta 3	ECM Smart	35 W	3.6 m³/h	6.9 m	Premium residential with digital display

Evoplus (Small)	ECM Com.	Up to 110 W	Up to 12 m³/h	Up to 11 m	Light commercial / multi-family buildings
Evoplus (Large)	ECM Com.	Up to 1.5+ kW	Up to 75 m³/h	Up to 18 m	Heavy commercial heating and cooling

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XV. LOWARA (A Xylem Brand)

Lowara is essentially the European sister-brand to Bell & Gossett (both owned by Xylem). While B&G dominates North America, Lowara is the badge you will see across Europe and Asia.

- **Architecture:** Shares the same spherical motor and ultra-efficient ECM technology as the B&G ecocirc lines.
- **Nomenclature: Ecocirc** = Standard ECM | **Ecocirc XL** = Commercial ECM | **N** = Stainless Steel.

Performance Matrix (Standard 230V, 50/60Hz)

Model	Type	Max Power	Max Flow	Max Head	Primary Application
Ecocirc (Basic/Premium)	ECM Smart	4 - 60 W	3.2 m³/h	6 m	Residential hydronic & underfloor heating
Ecocirc XL / XLplus	ECM Com.	Up to 1.5 kW	Up to 135 m³/h	Up to 18 m	Commercial boiler/chiller systems

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XVI. BIRAL

Biral is a Swiss manufacturer known for extreme precision, high build quality, and dominance in specific European and high-end OEM markets.

- **Architecture:** Highly distinctive control panels, often featuring glowing indicators. Built for maximum longevity and energy efficiency.
- **Nomenclature: ModulA** = Variable Speed Commercial | **PrimAX** = Residential | **A** = Standard / Legacy.

Performance Matrix (Standard 230V, 50/60Hz)

Model	Type	Max Power	Max Flow	Max Head	Primary Application
PrimAX	ECM Smart	Up to 50 W	Up to 4.5 m³/h	Up to 8 m	High-end residential hydronics
ModulA	ECM Com.	Up to 1.5 kW	Up to 70 m³/h	Up to 18 m	Premium commercial building management

EXPANDED MASTER FIELD REFERENCE: Circulators (Part 7 - Legacy German Acquisitions)

XXI. LOEWE PUMPENFABRIK (Acquired by Grundfos)

Loewe Pumpenfabrik (based in Lüneburg, Germany) was founded in 1872 and was one of the premier manufacturers of piston pumps, centrifugal pumps, and cast-iron heating circulators in Europe.

The History: In 1988, Grundfos acquired Loewe to solidify its dominance in the German industrial market. For a few years, they operated under the name "Grundfos Industrie-pumpen GmbH," but by 1996, the Loewe brand was officially retired.

Field Rule: If you find a dead Loewe pump in a basement boiler room today, you cannot buy Loewe parts. You must cross-reference the flange-to-flange dimensions and swap it for a modern **Grundfos** equivalent.

Legacy Loewe Heating Circulators & Modern Replacements (230V, 50Hz)

Legacy Loewe Model Line	Original Type	Modern Grundfos Replacement	Primary Application

Loewe Silenta (e.g., P233AD)	1-Speed / 3-Speed Cast Iron	**Grundfos UPS** or **ALPHA2**	Standard residential hydronic heating
Loewe CNP / CNL Series	Heavy-Duty Cast Iron	**Grundfos MAGNA1** or **MAGNA3**	Light commercial / multi-zone residential
Loewe Centritherm	Inline Boiler Pump	**Grundfos TP** Series	Older commercial boiler loops
Loewe Pomona	Multi-purpose / Transfer	**Grundfos NS** or **CMBE** Series	General water transfer / light boosting